MILITARY SPECIFICATION

COUPLER, MONITOR CU-737( )/URC

1. SCOPE

1.1 This specification covers the Coupler, Monitor CU-737( )/URC which is an equipment for ship or shore use. This equipment matches a 50 ohm output from an RF amplifier to a 50 ohm transmission line having a standing-wave ratio up to 2 to 1. The operating frequency of the unit is from 2.0 to 30 mc. This equipment contains an antenna transfer relay, a directional coupler with an RF wattmeter and a reversible L-network for impedance matching. A 4 ohm loudspeaker is provided as well as a terminating load which is switched to the audio input when the loudspeaker is not in use. This equipment is normally used in antenna systems using antenna multicouplers, in direct 50 ohm antenna systems and in submarine installations using Antenna Tuning Group AN/BRA-3, 5 or equivalent.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction model. - Preproduction model shall be furnished for the preproduction inspection specified in 4.3. The model shall be of the production type.

3.2 Materials and parts. - The materials and parts for the equipment shall be of the same quality, grade, type, characteristics, size, weight as the material used in the Government furnished equipment (GFE) (see 3.8).

3.3 Electrical design. - Unless otherwise specifically approved by the bureau concerned, the electrical design of the equipment shall be the same as that of the GFE, including all authorized equipment field changes that may or may not be in the furnished model.

3.4 Construction. - The construction of the equipment shall be the same as that of the GFE.

3.5 Parts interchangeability. - All parts shall be completely mechanically, electrically and functionally interchangeable with the corresponding parts as used in the GFE.

3.6 Performance. - The performance shall be within the limits specified herein. In the event of any conflict or inconsistency between the performance or other requirements of this specification and the GFE, this specification shall govern. It shall be the responsibility of the contractor to determine whether any such conflicts or inconsistency exist, and the contractor shall immediately, upon discovering any such conflict or inconsistency, notify the bureau concerned in writing, giving full details.

3.7 Operating conditions. - The equipment shall be capable of operation in accordance with the requirements of this specification under the following conditions:

(a) Ambient temperature range: -10°C. to +50°C.
(b) Ambient humidity range: 0 to 90 percent relative humidity.

3.7.1 Continuous power output. - The equipment shall be capable of withstanding a continuous power output of 500 watts average power.

3.7.2 Frequency range. - The frequency range of the equipment shall be continuous 2.0 to 30 megacycles (mc.).

3.7.3 Tuning capabilities. - The equipment shall be capable of tuning a 2 to 1 standing wave to at least a 1.1 to 1 standing wave over the entire 2.0 to 30 mc. frequency range.

3.7.4 Speaker. - The speaker shall be 4 ohms, with a capacity of 2 watts.

3.7.5 Characteristic impedance. - The characteristic impedance shall be 52 ohm, input and output.

3.8 Government furnished equipment (GFE). - Equipment models will be furnished by the Government. The contractor shall conduct all necessary tests of the GFE to confirm mechanical and electrical

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conformance to or in excess of these specification requirements within 90 days after receipt. The contractor shall notify the bureau concerned in writing, within 90 days after receipt of the GFE, and prior to the assembly of a preproduction model that the GFE model meets or exceeds these specifications, and that the preproduction and production units to be delivered will conform in all respects.

3.9 Workmanship. - Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination and tests. - Examination and tests performed on the equipment shall be in accordance with Specification MIL-E-16400, except as otherwise specified herein. Examination and tests shall consist of the following:

(a) Contractors inspection.
(b) Preproduction inspection (see 4.3).
(c) Contractors production line tests and examination (see 4.4).
(d) Acceptance inspection, including production and production control inspection (see 4.5).

4.3 Preproduction inspection. - Preproduction inspection shall be performed at the manufacturer's plant, under the supervision of the Government inspector, on the preproduction equipment offered for delivery. Inspection shall be in accordance with the following examination and tests of Specification MIL-E-16400:

(a) Surface examination.
(b) Operating test.
(c) Weights and dimensions.
(d) Supply line voltage and frequency.
(e) Heat test.
(f) Temperature.
(g) Humidity.
(h) Power.
(i) Controls and control circuits.

Prior to the delivery of any production units, the contractor shall receive an approval for production from the bureau concerned. Approval will be granted upon satisfactory completion of the preproduction inspection and the satisfactory modification of any defects or nonconformance with the requirements of this specification. The preproduction inspection report shall be submitted to the bureau concerned.

4.4 Contractors production line test and examination. - Every production equipment shall be subjected to a production line final test and examination by the manufacturer. In addition to a general inspection the adjustments, test requirements, and test limits specified in 4.4.1 through 4.4.4 shall be accomplished.

4.4.1 Calibration of directional coupler. - The directional coupler unit within every CU-737( )/URC equipment shall be initially calibrated prior to the overall tests in accordance with the procedures outlined in the CU-737( )/URC technical manual NAVSHIPS 93628, (GFE).

4.4.2 Adjustments. - Adjustments shall be made as follows:

(a) Remove side plate and variable capacitor knob on line tuner section of the CU-737( )/URC.
(b) Turn variable inductance fully counterclockwise to end stop and set knob to "O" reading at index.

4.4.3 Test requirements. - Test requirements shall be as specified in 4.4.3.1 through 4.4.3.3.

4.4.3.1 Low impedance operation. - Low impedance operation shall be conducted as follows:

(a) Connect output connector J3 to a 25 ohm dummy load, using 52 ohm coaxial cable. Set variable inductance at a setting of 600. Place impedance switch on "LO". Set power output switch to "Forward 1000". Provide a 500 watt average power 52 ohm source and tune to 2.0 mc. tune power source equipment until a reading of 500 watts is indicated on the CU-737( )/URC wattmeter. Place power output switch to "Reflected 100" position and readjust the variable inductance and variable capacitance controls of the CU-737 ( )/URC for minimum reflected power. Record reflected power reading and final reading of the variable inductance and capacitor dials.

4.4.3.2 High impedance operation. - High impedance operation shall be conducted as follows: Connect output connector J3 to a 52 ohm dummy load, using 52 ohm coaxial cable. Place the

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impedance switch to the "HI" position, set the variable capacitor to "0" and the variable inductance to "0". Set power output switch to "Forward 1000" position. Provide a 500 watt average power, 52 ohm source and tune to 2.0 mc. Tune power source equipment until a reading of 500 watts indicated on the CU-737( )/URC wattmeter. Place power output switch to "Reflected 100" position and readjust the variable inductance and variable capacitor controls of the CU-737( )/URC for minimum reflected power. Record the reflected power reading and final reading of the variable inductance and capacitor dials.

4.4.3.3 Low impedance and high impedance operation. - Low impedance and high impedance operation shall be conducted as follows:

(a) Reflected power shall be not more than 2 watts.
(b) Variable inductance dial setting shall be 600 ± 150.
(c) Variable capacitor dial setting shall be between 90 and 100.
(d) Record equipment serial number.

4.4.4 Unsatisfactory completion of this final production line test shall be cause for rejection. The contractor shall retain copies of each equipment final test report for a period of five years after delivery of the final equipment and make such reports available to an authorized Government representative upon request.

4.5 Acceptance inspection. - Acceptance inspection shall consist of the inspection specified in 4.5.1 and 4.5.2.

4.5.1 Production inspection. - Production inspection shall be conducted on every equipment offered for delivery in accordance with the following examination and tests specified in Specification MIL-E-16400, except as otherwise specified herein:

(a) Surface examination.
(b) Operating test.

4.5.2 Production control inspection. -

4.5.2.1 Samples for production control inspection. - Sample equipment shall be selected as follows for the inspection specified in 4.5.3:

(a) First 10 production units - 4 units shall be tested.
(b) Next 10 production units - 2 units shall be tested.
(c) Following production units to end of contract - On one unit of every 20 units.

4.5.3 Production control inspection shall be conducted in accordance with the following examination and tests specified in Specification MIL-E-16400, except as otherwise specified herein:

(a) Weights and dimensions.
(b) Heat test.
(c) Controls and control circuits.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - The equipment shall be preserved and packaged by level A or C; packed by level A, B or C and marked as specified (see 6.1); in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging and packing requirements (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
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